Microelectronics Journal 152 (2024) 106375

Contents lists available at ScienceDirect

Microelectronics Journal

journal homepage: www.elsevier.com/locate/mejo

A chopper instrumentation amplifier with discrete-time compensation based on current generation unit to eliminate electrode DC offset

Zichao Wang [a], Kui Wen [a], Ruixue Ding [a,b,*], Shubin Liu [a], Zhangming Zhu [a,b]

[a] Key Laboratory of Analog Integrated Circuits and Systems (Ministry of Education), School of Integrated Circuits, Xidian University, Xi'an 710071, China
[b] Hangzhou Institute of Technology, Xidian University, Hangzhou 311200, China

ARTICLE INFO

Keywords:
Bio-potential signals
Chopper stabilization
Electrode DC offset
Capacitively-coupled chopper instrumentation amplifier
Discrete-time compensation
Current generation unit

ABSTRACT

A capacitively-coupled chopper instrumentation amplifier (CCIA) for bio-potential signals acquisition is proposed in this paper. A novel discrete-time compensation scheme based on the current generation unit is adopted to suppress the DC offset caused by the sampling electrode. Different with the traditional analog DC servo loop (DSL) used in previous works, the circuit based on this scheme is more straightforward and consumes less power. Moreover, it can suppress a larger electrode DC offset in a short time. This work is designed in a standard 180 nm CMOS process. The CCIA operates from a 1.8 V supply, from which it draws a total current of 0.72 μA. The simulation result shows that the signal bandwidth of the proposed CCIA is 1.2 – 500 Hz and the mid-band gain is about 31.49 dB. In the frequency band of 1 – 500 Hz, the input-referred noise of the circuit is 2.01 μV_{rms}. Inputting a single-tone sine wave with an amplitude of 14.1 mV at the frequency of 56.152 Hz, the total harmonic distortion (THD) of the CCIA's output is −51.38 dB. This circuit can suppress a large electrode DC offset within a few milliseconds, and the maximum electrode DC offset that can be tolerated up to 130 mV.

1. Introduction

Bio-potential signals such as electrocardiograph (ECG) can reflect the health status of people and provide critical information for medical institutions to diagnose patients [1,2]. By collecting bio-potential signals, new medical treatment methods such as telemedicine and innovative hospitals provide new solutions to problems such as maldistribution of medical resources, difficulties of patients seeking medical treatment, and high operating pressure of hospitals [3]. However, bio-potential signals of the human body are characterized by small amplitude, low frequency, narrow bandwidth and numerous in-band interferences [4,5]. When acquiring these signals, the main disturbances include the thermal and 1/f noise of the circuits, 50/60 Hz power line interference, motion artifacts, and DC offset caused by the sampling electrodes [6,7]. Different with other noise, the electrode DC offset cannot be eliminated by auto-zero (AZ), chopper stabilization (CHS), or other techniques [5]. Different electrodes generate different half-cell potentials on skin surface, which results in an offset voltage at the input of the circuit [8]. Generally, commercial wet electrodes produce a DC offset of about 50 mV [9]. If the offset is not suppressed correctly, it will saturate the AFE circuit.

In previous works, off-chip components were used as the input stage to suppress the electrode DC offset. However, it reduced the integration of the circuit [10]. AC-coupled input stage has also been used in some papers [11]. Although the pseudo resistor with high resistance is simple in structure, it is easily affected by process, power, or temperature, which brings great nonlinearity to the circuit. Recently, a negative feedback DC servo loop (DSL) based on active integrator is usually integrated into the AFE, the core of which is to provide a low-frequency high-pass corner frequency [12]. Nevertheless, it requires enormous resistors and capacitors, increasing the area overhead. A DSL based on duty-cycle resistors (DCR) is reported in [13,14]. DCR can control the resistance by adjusting the clock duty cycle. However, as its resistance is susceptible to clock accuracy, its practicality could be better. In addition, these DSLs require a long time to integrate the amplifier's output. As a result, it takes a long time to respond to the electrode DC offset. Moreover, due to the limitation of the operational transconductance amplifier (OTA)'s output swing, the maximum electrode DC offset that can be suppressed is limited. Therefore, this paper proposes a discrete-time compensation scheme based on the current generation unit. Compared with the DSLs mentioned above, it has a

* Corresponding author at: Key Laboratory of Analog Integrated Circuits and Systems (Ministry of Education), School of Integrated Circuits, Xidian University, Xi'an 710071, China.
E-mail address: rxding@mail.xidian.edu.cn (R. Ding).

https://doi.org/10.1016/j.mejo.2024.106375
Received 12 July 2024; Received in revised form 17 August 2024; Accepted 18 August 2024
Available online 22 August 2024

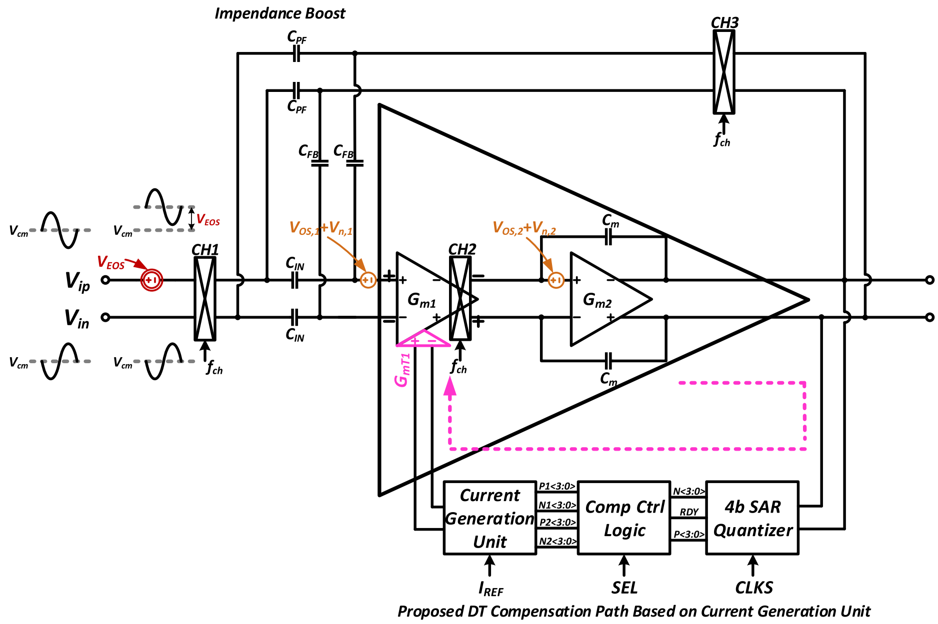

Fig. 1. Complete implementation of the proposed AFE amplifier with discrete-time compensation for acquiring bio-potential signals.

simpler structure, smaller area, and faster response, and it can suppress a larger DC offset.

This paper is organized as follows: In Section 2, the architecture and compensation process of the AFE with the proposed discrete-time compensation are introduced. In Section 3, the functions and specific implementations of the critical circuit are introduced and analyzed in detail. The simulation results are presented in Section 4. Finally, the conclusion is given in Section 5.

2. Architecture of overall circuit and workflow of the compensation path

As shown in Fig. 1, the AFE circuit used to collect bio-potential signals in this paper adopts a capacitively-coupled chopper instrumentation amplifier (CCIA) topology, which is mainly composed of a two-stage OTA, a capacitive feedback network (input capacitor C_{IN} and feedback capacitor C_{FB}), input modulation chopper CH1, output demodulation chopper CH2, and feedback modulation chopper CH3. The proposed CCIA can amplify the input signals while suppressing the offset and 1/f noise of the circuit. Ignoring the mismatch of capacitors in the differential circuit and the finite open-loop gain of the OTA, the voltage gain A_{CCIA} of the CCIA is given by

$$A_{CCIA} = \frac{C_{IN}}{C_{FB}} \tag{1}$$

The core of the CCIA is a two-stage OTA with Miller compensation, marked as G_{m1} and G_{m2} in Fig. 1. CH1 is placed before C_{IN} to suppress the offset caused by the mismatch of the capacitors. CH2 is embedded in the first-stage OTA, G_{m1}. With the use of CHS, the offset and 1/f noise of G_{m1} can be modulated to the high-frequency harmonics of the chopping frequency and then filtered out by the integrator built around G_{m2}, thereby improving the signal-to-noise ratio (SNR) of the circuit. The gain of G_{m1} suppresses the offset and 1/f noise of G_{m2}. Pseudo resistor connected to the input of the OTA (not shown in Fig. 1) provides the correct bias to the CCIA.

The capacitively-coupled structure offers a large DC input impedance, and by adjusting the input capacitance to less than 100 pF, the attenuation of the signal at the electrodes is negligible. However, in this work, a chopper design is used to modulate the DC and low-frequency signals to high frequency. Then the signal is capacitively coupled to the input of the OTA, decreasing the DC input impedance of the circuit. The combination of chopper and capacitor forms a switched-capacitor resistor, and its equivalent resistance R_{eq} can be expressed as follows.

$$R_{eq} = \frac{1}{2\, f_{chop}\, C_{IN}} \tag{2}$$

where f_{chop} is the chopping frequency.

To avoid the attenuation of the signal at the electrodes and the damage to the tissue by large currents, a positive feedback loop (PFL) is adopted. As shown in Fig. 2, it connects the CCIA's input and output via CH3 and C_{PF}. The PFL converts the output voltage into current, and then feeds it back to the input of the CCIA to compensate for the current drawn by the feedback branch; thereby enhancing the DC input impedance of the circuit. Ideally, when the current I_{PF} on the PFL is equal to the current I_{FB} on the feedback path, the current that CCIA drawn from the source is 0, and the DC input impedance of the circuit is infinite.

$$I_{PF} = 2(V_{out} - V_{in}) f_{chop} C_{PF} = 2V_{out} f_{chop} C_{FB} = I_{FB} \tag{3}$$

As shown in Fig. 1, different from the circuit's own noise, the DC offset from the sampling electrodes is modulated to high frequency along with the signal by CH1 and then coupled to the input of G_{m1} via C_{IN}. Previous works use a DSL based on the active RC integrator to suppress the electrode DC offset and avoid saturation of the CCIA. Considering that the bio-potential signal has the characteristics of low frequency and narrow bandwidth, the DSL based on the RC integrator needs to provide a very low high-pass corner frequency for the AFE circuit, which can attenuate the electrode DC offset while retaining the bio-potential signal. This method requires a large resistor, which increases the area overhead. Meantime, traditional DSL has a slower response to the offset, and it takes a long time to restore the output voltage.

In this paper, a discrete-time compensation scheme based on the current generation unit is proposed. Compared with the traditional analog DSLs, it has a more straightforward structure and smaller area, which can quickly suppress the large DC offset caused by the sampling electrode of the AFE circuit so that the amplifier works in a proper bias state and avoids output saturation. The implementation of the proposed discrete-time compensation circuit is shown in Fig. 3. It comprises the CCIA above, a 4-bit SAR quantizer, a compensation control logic, and

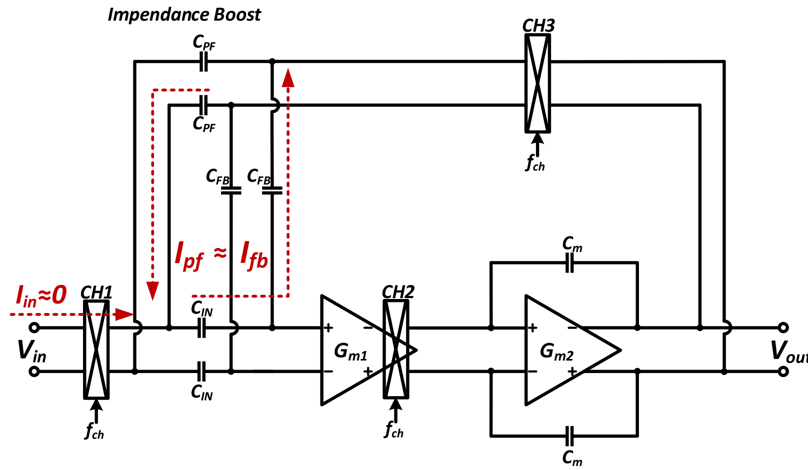

Fig. 2. A schematic of the PFL used to improve the CCIA's DC input impedance.

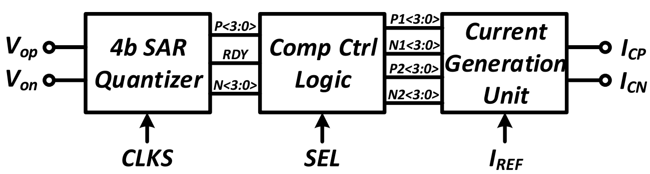

Fig. 3. The generation process of the compensation current I_{CP} and I_{CN}.

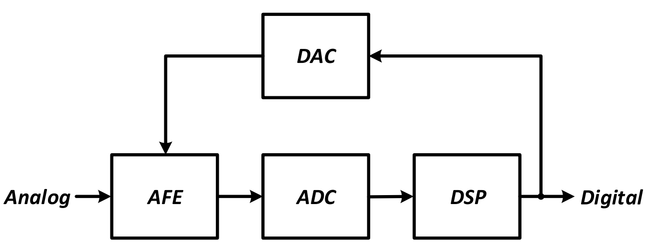

Fig. 4. Diagram of signal acquisition and processing system.

a 4-bit binary-weighted current generation unit. Fig. 3 illustrates the generation process of the compensation current I_{CP} and I_{CN}.

The specific workflow of the compensation path is as follows:
(1) Under the control of the sampling clock CLKS, the 4-bit SAR quantizer samples and quantizes the electrode DC offset of the CCIA's output. After all the binary codes are quantized, they are stored in the compensation control logic.
(2) The compensation control logic sends these codes to the current generation unit to control the closure of the corresponding weighted switches, which then generates a suitable compensation current.
(3) The current feeds back to G_{m1} and compensates the current variation caused by the voltage change at the input. This maintains a stable current in G_{m1}, thereby avoiding the saturation of the CCIA.

3. Critical circuit and its implementation in compensation path

The critical circuits of the proposed discrete-time compensation path include a 4-bit asynchronous SAR quantizer, a compensation control logic, and a 4-bit binary-weighted current generation unit. This section will introduce and analyze these circuits in detail.

3.1. 4-Bit SAR quantizer

As shown in Fig. 4, the analog-to-digital converter (ADC) can convert analog signals into digital signals, bridging the AFE system and the digital signal processing (DSP) system. This design adopts a successive approximation register (SAR) quantizer, which is used to quantize the analog voltage at the CCIA's output into a corresponding 4-bit binary code.

The accuracy of the discrete-time compensation scheme is related to the precision of the SAR quantizer, the residual DC offset at the CCIA's input is less than $V_{LSB}/(2A_{CCIA})$, where V_{LSB} is the minimum quantization step of the SAR quantizer. However, too many bits in the SAR quantizer will increase the requirement of the reference current and make the layout of the binary-weighted current generation unit more difficult. Overall, the SAR quantizer precision is set to 4-bit, and the residual DC offset voltage at CCIA's input is less than 8 mV.

Fig. 5 shows the 4-bit asynchronous SAR quantizer used in this design, which includes a bootstrapped switch, a capacitive DAC (CDAC) array, a comparator, a SAR control logic, and an asynchronous clock generator. To improve the common mode rejection ratio (CMRR) of the circuit, the SAR quantizer adopts a fully differential structure. Moreover, considering the lower precision requirements for the quantizer in this work, the quantizer employs V_{CM}-Based switching timing, which reduces the switching energy and improves the conversion linearity [15]. The workflow of the quantizer can be divided into two phases: sampling and conversion. During the sampling phase, the input signal is sampled by the CDAC array through the bootstrapped switches. During the conversion phase, the differential voltage on the CDAC is fed into the comparator for comparison. Based on the comparison results, a binary code is generated and stored for subsequent output. At the same time, this code will also control the closure of the switches in the CDAC array to generate a correct differential voltage for the successive approximation logic to conduct the following comparison. Finally, under the control of the comparison clock CLKC, the comparator generates four binary codes, and the analog input is converted to a digital output.

The bootstrapped switches are controlled by the sampling clock CLKS. When CLKS is high, the sampling transistor turns on and the input signal is followed. When CLKS is low, the sampling transistor turns off and the input signal is sampled onto the CDAC array. The bootstrapped switches reduce the variation of the sampling transistor's on-resistance by keeping the gate-source voltage stable, thus reducing the signal distortion caused by sampling.

This paper adopts a low noise and low offset double-tail two-stage fully dynamic comparator. Fig. 6 shows its details at the transistor level. The first stage is the dynamic pre-amplification (pre-amp) stage and the second stage is the latch with positive feedback. Compared with the traditional comparator, this double-tail two-stage dynamic comparator has fewer stacked transistors and is more suitable for low-voltage process design. The double-tail structure can achieve fast latching independent of the input common-mode (CM) level by passing

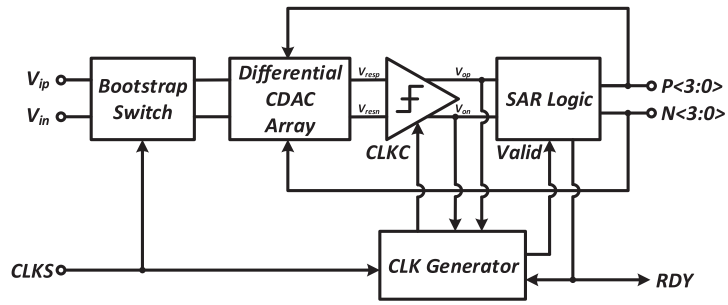

Fig. 5. Block diagram of the proposed 4-bit asynchronous SAR quantizer.

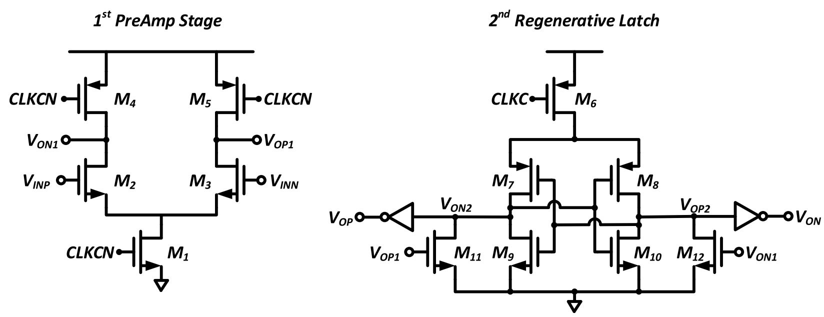

Fig. 6. Schematic of the double-tail two-stage dynamic comparator at the transistor level.

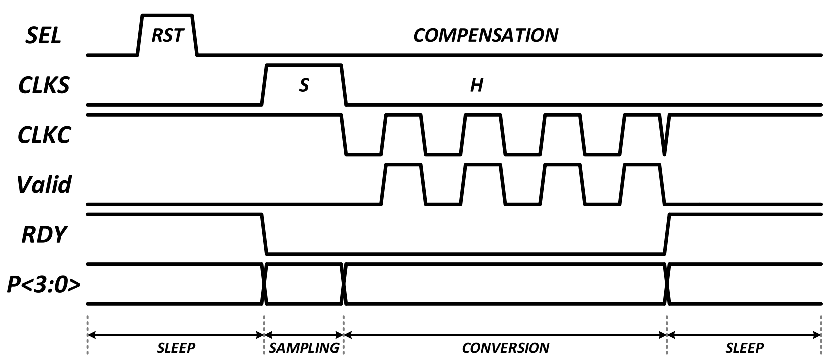

Fig. 7. Timing diagram of the 4-bit asynchronous SAR quantizer.

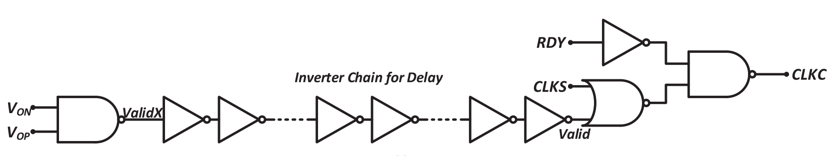

Fig. 8. The implementation of asynchronous clock generator.

a large current through the second stage, and a low input offset by allowing the pre-amp stage to flow a low current. When CLKC is high, the dynamic comparator is in the set phase, and when CLKC is low, the dynamic comparator works in the comparison phase. In addition, the use of M11 and M12 also isolates the input and output, reducing the kickback noise of the dynamic comparator.

The working timing of the SAR quantizer is shown in Fig. 7, where SEL is an enable signal, which will be introduced in the next section. CLKS is the sampling clock used to control the bootstrapped switches. CLKC is the asynchronous comparison clock of the comparator. Valid is the flag signal of the comparator, and RDY is the conversion completion signal of the SAR quantizer. The implementation of the asynchronous clock generator is shown in Fig. 8.

3.2. Compensation control logic

The compensation control logic is used to control the current generation unit to compensate the electrode DC offset of the CCIA, and its structure is shown in Fig. 9. Under the control of the enable signal SEL, the workflow of the compensation control logic can be divided into two stages. In the first stage, it outputs the set signal and is disable from compensating. The CCIA amplifies the electrode DC offset directly. In the second stage, the SAR quantizer samples the offset and converts it into a 4-bit binary code. Then, according to the binary code, the compensation control logic generates a corresponding logic signal, which controls the current generation unit to generate a proper compensation current.

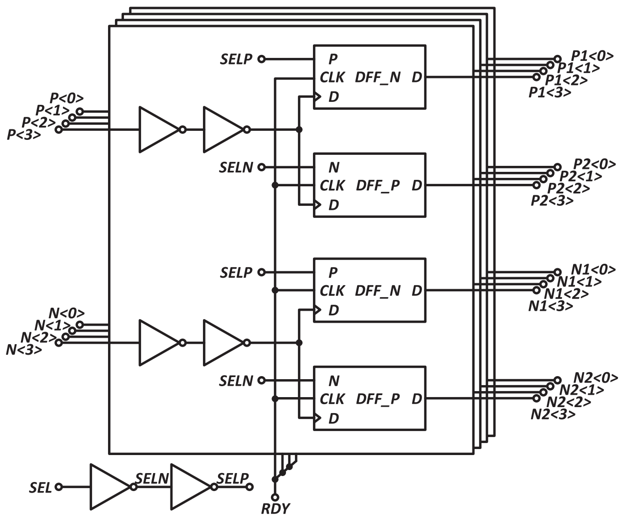

Fig. 9. Schematic diagram of the compensation control logic.

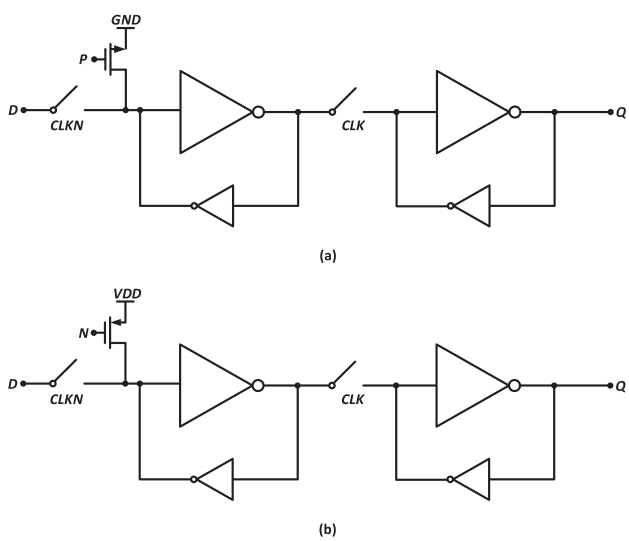

Fig. 10. (a) DFF with NMOS (b) DFF with PMOS.

The compensation control logic consists of four identical subunits, each of which includes an inverter, a D flip-flop (DFF) with PMOS, and a DFF with NMOS. Fig. 10 shows the schematic of the DFF with PMOS and the DFF with NMOS, both of which adopt a master–slave flip-flop structure. The former will set the output to VDD with a p-type transistor, while the latter will set the output to GND with an n-type transistor.

The control signal of the setting transistor is generated by SEL. The relative timing between SEL and other signals in the SAR quantizer mentioned is shown in Fig. 7. The pulse of SEL occurs before the high level of CLKS, in the meantime, RDY is also high. When SEL is high, the compensation control logic outputs a set signal, and the current generation unit is disabled from compensating. When the pulse of SEL ends, the setting transistor of the DFF turns off, and the compensation path starts to work.

3.3. Binary-weighted current generation unit

Considering the lower resolution of DAC required in this design, a binary-weighted current generation unit is used. This type of circuit has a simple structure, fast conversion, and a few current source. Moreover,

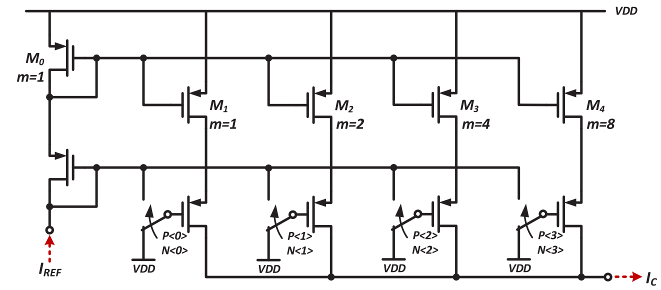

Fig. 11. Schematic of the proposed binary weighted current generation unit at the transistor level.

as the previous quantizer outputs the binary code, compared with the thermometer coding current generation unit, it is unnecessary to use an additional binary-to-thermometer decoding circuit, greatly reducing the layout area of the digital part.

The specific implementation of the current generation unit is shown in Fig. 11. A cascode current mirror is used to suppress the effect of channel-length modulation. The dimensions ratio of the current mirror transistors M1-4 to the diode-connected transistor M0 is 1:2:4:8:1, and the corresponding binary-weighted currents are copied respectively. When SEL is high, the compensation logic outputs the set signal. Therefore, the branches, which are composed by the transistors M1-4, are cut off. When SEL is low, the compensation control logic controls the closure of the corresponding weighted switches in the current generation unit according to the binary code output by the quantizer. Finally, the current generation unit generates an appropriate compensation current and feeds it back to the CCIA.

3.4. Two-stage OTA

The core of the CCIA is a two-stage OTA, and its details at the transistor level are shown in Fig. 12 (the common mode feedback circuit is omitted). The first stage uses a folded-cascode OTA with PMOS input, which can provide a larger CM input range. The second stage uses a common-source stage with current-source load, increasing both the output swing and the gain of the OTA. Miller compensation is implemented to ensure the stability of the loop.

OTA are typically designed with larger input pairs to reduce the $1/f$ noise and enhance the matching of layout. Using 3–5 times the minimum channel length can effectively eliminate the influence of process errors on circuit design. However, increasing the channel length will increase the overdrive voltage of the transistor that works in saturation, thereby consuming more voltage margin, which is not conducive to circuit design under a low-voltage process. Therefore, in this design, the input transistor of the first stage adopts a self-cascode composite transistor structure with two PMOS transistors in series. Among them, the PMOS near the power supply operates in linear region, while the lower operates in saturation. Compared with the single PMOS as the input, the equivalent simple transistor of this structure increases the channel length. Meantime, as the upper PMOS works in linear region, its drain–source voltage is very small, and the increment of the overdrive voltage can be neglected nearly. Therefore, it is very suitable for the design of high-precision and power efficient circuits in the low-voltage process.

Node X in the first-stage OTA is the node where the compensation current feeds back. When an offset voltage feeds into the first-stage OTA, different DCs flow through the input pairs. As the current on transistor M5 and M6 is the sum of the current on the input and the cascode branch, the DCs flowing through the cascode branches are also different. Therefore, a large DC offset voltage occurs at the OTA's output and then transmits to the next stage. By adopting the proposed discrete-time compensation scheme, a proper compensation current

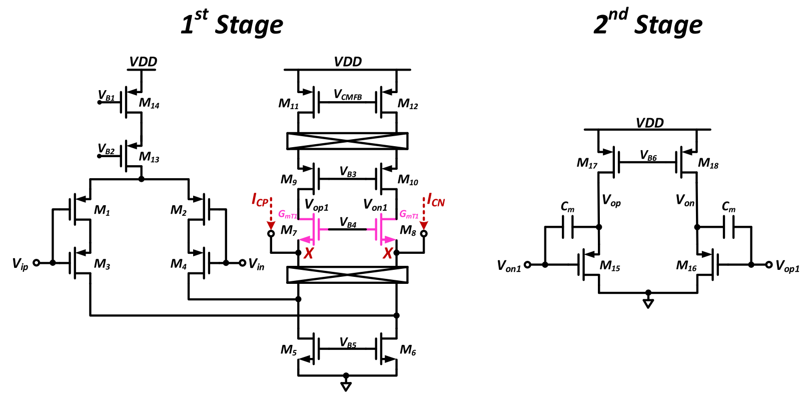

Fig. 12. Schematic of the two-stage operational amplifier at the transistor level.

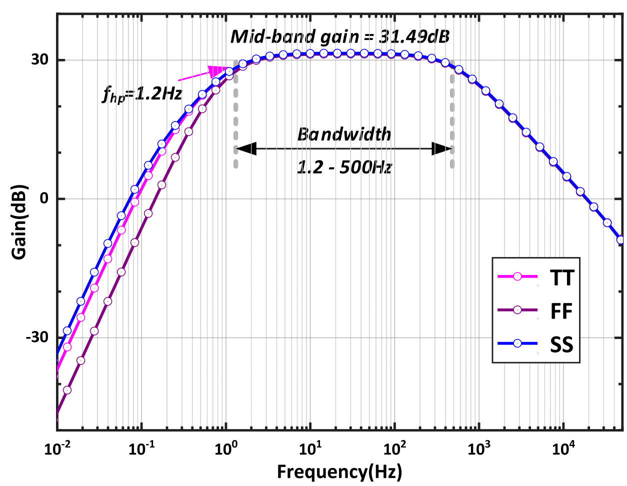

Fig. 13. Frequency response of the CCIA at the TT, FF, and SS process corners.

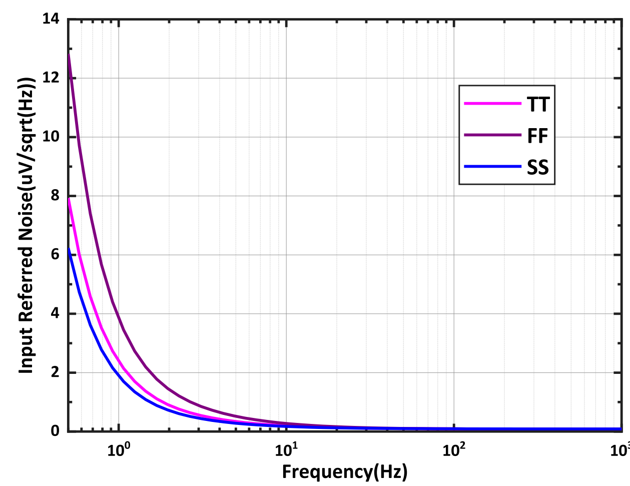

Fig. 14. The input reference noise spectral density of the CCIA at different process corners.

is generated and fed back to node X, ensuring that the DCs flowing through the cascode branches remain consistent and suppressing the impact of electrode DC offset. The reference current I_{REF} in the current generation unit can be determined as follows.

$$I_{REF} = \frac{1}{2} g_{m-eff} \frac{\frac{2V_{REF}}{2^N}}{A_{CCIA}} = \frac{1}{2} g_{m-eff} \frac{V_{LSB}}{A_{CCIA}} \quad (4)$$

where g_{m-eff} is the equivalent transconductance of the composite transistor and V_{LSB} is the minimum quantization step of the SAR quantizer.

4. Simulation results

The proposed CCIA with discrete-time compensation is implemented in a standard 180 nm CMOS process. The circuit operates at a 1.8-V supply with a total current of 0.72 μA. Fig. 13 shows the frequency response of the CCIA at the TT, FF, and SS process corners. As shown in Fig. 13, the mid-band gain of the CCIA is about 31.49 dB at all different process corners. The high-pass corner frequency at the TT process corner is 1.2 Hz, and the signal bandwidth is 500 Hz. Combined with the proposed discrete-time compensation scheme, the CCIA can effectively suppress the DC offset caused by the sampling electrode, and the maximum electrode offset voltage that can be tolerated up to 130 mV. The input-referred noise spectral density of the CCIA within 1 kHz bandwidth at different process corners is shown in Fig. 14. In the frequency band of 1–500 Hz, the integral noise at the TT, FF and SS process corners is 2.3 μV_{rms}, 3.28 μV_{rms} and 2.01 μV_{rms}, respectively.

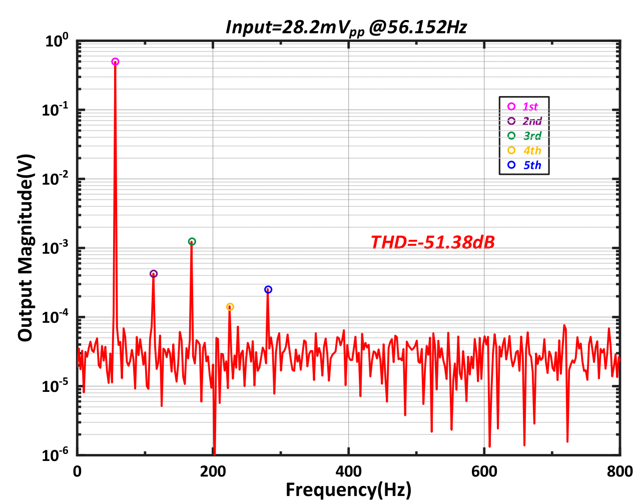

Fig. 15. The harmonic distortion of the proposed CCIA's output.

Fig. 15 shows the harmonic distortion of the proposed CCIA's output. The input signal is a single-tone sine wave with an amplitude of 14.1 mV at the frequency of 56.152 Hz. A 2048-points fast Fourier transform (FFT) is performed on the CCIA's output. Considering the first five harmonics (2nd, 3rd, 4th, and 5th), the total harmonic distortion (THD) of the CCIA is calculated as −51.38 dB, which is sufficient to

Table 1
Performance summary and comparison with prior works.

Reference	[16]	[17][a]	[18]	[9]	[19][a]	This Work
Topology	CCIA	CCIA	CCIA	CCIA	CCIA	**CCIA**
Process (nm)	28	180	180	180	180	**180**
Supply (V)	1	1.25	1.8	1.8	1.5	**1.8/5**
Current (μA)	0.65	1.7	2.02	7.63	1.2	**0.72**
Gain (dB)	25	34	26.06	39.15	37	**31.49**
Bandwidth (Hz)	0.51–4.4 k	0.5–10 k	0.1–5 k	0.5–10 k	0.8–300	**1.2–500**
Input-referred noise (μV_{rms})	4.65(0.51–4.4 kHz)	5.7(0.5–10 kHz)	0.58(1–200 Hz)	0.26(0.1–100 Hz)	11.8(1–300 Hz)	**2.01(1–500 Hz)**
Electrode DC offset Tolerance (mV)	20	5	50	50	N.A.	**130**
Input impedance (MΩ)	N.A.	1.8	958	N.A.	N.A.	**970**
NEF	2.7	2.9	2.32	2.76	9.9	**2.93**
PEF	7.29	10.52	9.68	13.72	147.02	**15.45**

[a] Measurement results.
N.A. stands for not available.

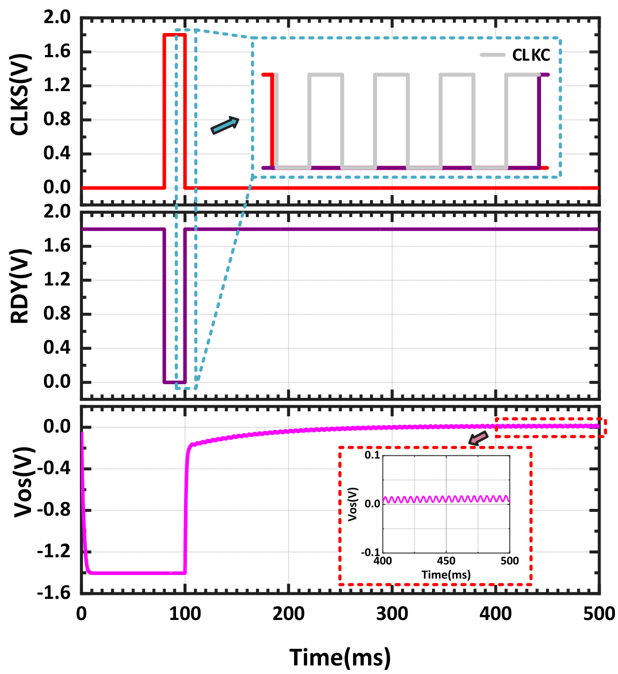

Fig. 16. The transient simulation of the circuit suppressing the electrode DC offset.

ensure that the AFE circuit can obtain the high-fidelity bio-potential signals.

Fig. 16 shows the transient simulation of the circuit suppressing the electrode DC offset. CLKS is the sampling clock, RDY is the conversion completion signal of the SAR quantizer, and V_{os} is the offset voltage at the circuit's output. When CLKS is high, the SAR quantizer samples the offset voltage at the CCIA's output, and RDY is low. When the CLKS is pulled down, the SAR quantizer quickly converts the sampled offset voltage into a 4-bit binary code, and then RDY pulls up, indicating the completion of the conversion. The generated binary code controls current generation unit to compensate the CCIA so that the output offset voltage can be suppressed quickly. As shown in Fig. 16, the offset voltage of 1.2 V is eliminated just in a few milliseconds.

The performance of the proposed CCIA with discrete-time compensation is summarized and compared with other state-of-the-art works in Table 1. Compared to other works, although the noise performance of this CCIA is limited, its power consumption is lower, and the total current required is only slight higher than [16]. As a result, it obtains a comparable noise efficiency factor (NEF) to other works, which is 2.93. In addition, the maximum electrode DC offset that the CCIA can tolerate is larger than other circuits, which equals to 130 mV, and as shown in Fig. 16, its elimination process of offset is also faster. Moreover, the proposed CCIA exhibits the highest DC input impedance, reaching an impressive 970 MΩ. This feature effectively mitigates signal attenuation on the electrode and minimizes potential tissue damage caused by high current. Fig. 17 shows the overall power consumption of the proposed circuit. It can be seen that the CCIA contributes the most to the power consumption.

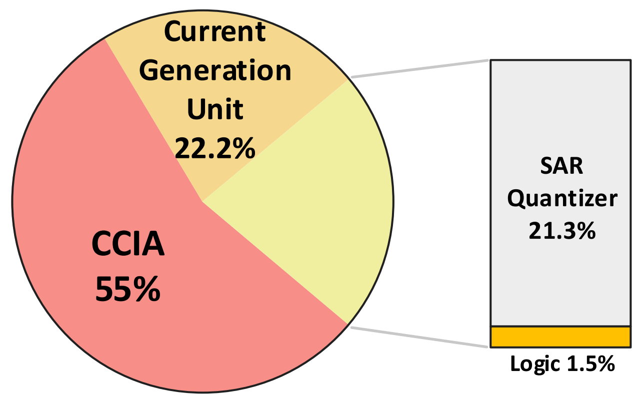

Fig. 17. The power breakdown diagram of the circuit.

5. Conclusion

In this work, a discrete-time compensation scheme based on the current generation unit is proposed to suppress the electrode DC offset in AFE circuits. Compared with the traditional analog DSLs, this circuit is more straightforward and consumes less power. Moreover, it can eliminate a larger electrode DC offset in a short time. The circuit is designed in a standard 180 nm CMOS process. The frequency response and input-referred noise spectral density of the CCIA are verified through corner simulations. This scheme can suppress a large electrode DC offset within a few milliseconds, and the maximum electrode DC offset that can be tolerated up to 130 mV.

CRediT authorship contribution statement

Zichao Wang: Methodology, Investigation, Writing – original draft & editing. **Kui Wen:** Conceptualization, Writing – review. **Ruixue Ding:** Data curation. **Shubin Liu:** Formal analysis. **Zhangming Zhu:** Visualization.

Declaration of competing interest

We wish to confirm the following facts that there are no known conflicts of interest associated with this manuscript, titled as "A chopper instrumentation amplifier with discrete-time compensation based on current generation unit to eliminate electrode DC offset", and there has been no significant financial and personal relationships with other people or organizations that can inappropriately influence the outcome of this work.

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

We confirm that we have given due consideration to the protection of intellectual property associated with this work, and that there are no impediments to publication, including the timing of publication, with respect to intellectual property. By doing this, we confirm that we have followed the regulations of our institutions concerning intellectual property.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China (62021004, 92164301), and in part by the Industry-University-Academy Cooperation Program of Xidian University-Chongqing IC Innovation Research Institute, China (No. CQIRI-CXYHT-2022-02, No. CQIRI-CXYHT-2022-04). All authors have read and agreed to publish the manuscript.

References

[1] Zhaoyang Ge, Huiqing Cheng, Zhuang Tong, Ziyang He, Adi Alhudhaif, Kemal Polat, Mingliang Xu, A knowledge-driven graph convolutional network for abnormal electrocardiogram diagnosis, Knowl.-Based Syst. 296 (2024) 111906.

[2] Haotian Yu, Yaguang Yang, Daibo Zhang, Qiliang Zhang, Zhiqiang Li, VLSI design for adjustable compression rate in lossless/lossy compression of EEG signal, Microelectron. J. 148 (2024) 106193.

[3] Z Khoshkam, A. Abrishamifar, A very low-power discrete-time delta-sigma modulator for wireless body area network, Microelectron. J. 130 (2022) 105633.

[4] Mehrnoosh Vafaei, Zahra Rahiminejad, Ebrahim Abiri, Mohammad Reza Salehi, A dual-output switched-capacitor regulator with a self-controlled ripple reduction technique for self-powered EEG acquisition, Microelectron. J. 135 (2023) 105758.

[5] Wentao Han, Qi Yu, Kejun Wu, Zhong Zhang, Jing Li, Ning Ning, A programmable analog front-end with independent biasing technique for ECG signal acquisition, Microelectron. J. 136 (2023) 105792.

[6] Yuyuan Tian, Lianxi Liu, Jian Ma, A high-sensitivity, low-noise dual-band RF energy harvesting and managing system for wireless bio-potential acquisition, Microelectron. J. 116 (2021) 105239.

[7] Maryam Gharaei Jomehei, Samad Sheikhaei, A low-power low-noise CMOS bio-potential amplifier for multi-channel neural recording with active DC-rejection and current sharing, Microelectron. J. 83 (2019).

[8] M.F Carlino, G. Gielen, A servo-loop-free charge sharing technique to mitigate electrode offsets in biomedical multiplexed interfaces, in: 2022 29th ICECS, Glasgow, United Kingdom, 2022, pp. 1–4.

[9] Tingyu Wang, Kui Wen, Longjie Zhong, Shubin Liu, A 0.26 μ V_{rms} instrumentation amplifier based on discrete-time DC servo loop with successive-approximation-compensation technique and Chopper DAC Array, Microelectron. J. 139 (2023) 105906.

[10] Y Perelman, R. Ginosar, An integrated system for multichannel neuronal recording with Spike/LFP separation, integrated A/D conversion and threshold detection, IEEE Trans. Biomed. Eng. 54 (1) (2007) 130–137.

[11] H Kassiri, et al., Battery-less tri-band-radio neuro-monitor and responsive neurostimulator for diagnostics and treatment of neurological disorders, IEEE J. Solid-State Circuits 51 (5) (2016) 1274–1289.

[12] Q Fan, F Sebastiano, J.H Huijsing, K.A.A. Makinwa, A 1.8 μW 60 nV / $\sqrt{Hz}$ capacitively-coupled chopper instrumentation amplifier in 65 nm CMOS for wireless sensor nodes, IEEE J. Solid-State Circuits 46 (7) (2011) 1534–1543.

[13] H Chandrakumar, D. Marković, A high dynamic-range neural recording chopper amplifier for simultaneous neural recording and stimulation, IEEE J. Solid-State Circuits 52 (3) (2017) 645–656.

[14] H Chandrakumar, D. Marković, An 80-mVpp linear-input range, 1.6-gΩ input impedance, low-power chopper amplifier for closed-loop neural recording that is tolerant to 650-mpp common-mode interference, IEEE J. Solid-State Circuits 52 (11) (2017) 2811–2828.

[15] Y Zhu, et al., A 10-bit 100-MS/s reference-free SAR ADC in 90 nm CMOS, IEEE J. Solid-State Circuits 45 (6) (2010) 1111–1121.

[16] Y Jiang, W Zuo, J Zheng, L. Cheng, A 0.65 μW 70nV/$\sqrt{Hz}$ capacitively-coupled chopper instrumentation amplifier for bio-potential sensing in 28-nm CMOS, in: 2020 IEEE International Conference on Integrated Circuits, Technologies and Applications, ICTA, Nanjing, China, 2020, pp. 47–48.

[17] J Wu, M.-K Law, P.-I Mak, R.P. Martins, A 2-μW 45-nV/$\sqrt{Hz}$ readout front end with multiple-chopping active-high-pass ripple reduction loop and pseudofeedback DC servo loop, IEEE Trans. Circuits Syst. II 63 (4) (2016) 351–355.

[18] F Ansari, M. Yavari, A high input impedance fully-differential chopper amplifier for closed-loop neural recording, in: 2020 28th Iranian Conference on Electrical Engineering, ICEE, Tabriz, Iran, 2020, pp. 1–5.

[19] Shashank Dwivedi, Anup K. Gogoi, A compact sub-hertz local field potential amplifier for implantable biomedical devices, Microelectron. J. 128 (2022) 105539.